

Chemistry Ch 5.1: Why Do Atoms React?

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Ek sawal se shuru karte hain. Sodium paani mein daalein to aag pakar leta hai. Chlorine zehreeli gas hai. Magar **helium** ko aap kuch bhi kar lein, wo kisi se nahin milta. Kyun? Ek element itna be-niyaz kyun hai?

Jawab: bhari hui bairooni shell

Noble gases ki bairooni shell **poori bhari** hoti hai. Wo pehle hi **mustahkam (stable)** hain, is liye unhein kuch karne ki zaroorat hi nahin.

Aur baqi tamam atoms **yehi halat** hasil karna chahte hain. Isi khwahish ka naam **chemical bonding** hai.

Do usool, aur dono yaad rakhne hain

Octet rule: atom apni bairooni shell mein **8 electrons** chahta hai.

Duplet rule: magar kuch chhote atoms ke liye **2 hi kaafi** hain, kyunke un ki pehli shell mein gunjaish hi 2 ki hai. Ye hain **hydrogen, lithium aur beryllium**.

Sawal mein aksar poochha jata hai ke octet aur duplet mein farq kya hai. Jawab: tadaad ka, 8 banam 2, aur kaun se atoms par lagu hota hai.

Teen raaste, aur teen qism ke bond

Atom mustahkam hone ke liye teen mein se ek kaam karta hai:

1. Electron de deta hai → cation banta hai

2. Electron le leta hai → anion banta hai

3. Electron share kar leta hai → covalent bond banta hai

Pehle do mil kar **ionic bond** banate hain. Teesra **covalent**. Baqi poora chapter inhi teen raaston ki tafseel hai.

Quiz

1. Why do noble gases not react? State the **octet rule** and the **duplet rule**, and name the three elements the duplet rule applies to.
2. Name the **three** ways an atom can achieve a stable outer shell, and say which type of bond each one leads to.